

29/6/25

HW-3 Answers

1) Given:

$\Rightarrow A$ is a $m \times d$ matrix

$\Rightarrow m \gg d$; $X = AA^T$

which makes X is a $m \times m$ matrix
where m is very high &
 d is small

$\Rightarrow X$ has d distinct, non-zero eigenvalues

To do:

$\Rightarrow$ Find eigen decomposition of X .

But, since $m \gg d$, finding eigen decomposition
is computationally expensive.

$\Rightarrow$ So, to find more efficient algorithm to
find eigen decomposition using $d \times d$ matrix,
which is smaller (Note: Not to use SVD
as a black box)

Solution:

$\Rightarrow$ As the non-zero eigenvalues will be
same for the $m \times m$ matrix ($X = AA^T$)
& the $d \times d$ matrix ($A^T A$), & the
eigen vectors of X can be reconstructed
using A & the eigenvectors of $A^T A$.
 $\Rightarrow$ This arises from PCA & spectral theory.

Let $B = A^T A$ ($d \times d$ matrix)

$\Rightarrow$ Eigen decomposition of B ,

$$B = V \Lambda V^T$$

where $V \Rightarrow$ eigen vectors of B

$\Lambda \Rightarrow$ diagonal matrix of eigenvalues of B

Then, Eigenvalues of $X = AA^T$ are

$\Rightarrow$ Same d non-zero eigenvalues $\lambda_1, \dots, \lambda_d$
as in Λ

$\Rightarrow$ Each eigenvalue is distinct

$\Rightarrow$ The rest $m-d$ eigenvalues are zero.

The corresponding eigenvectors u_i of R are

$$u_i = \frac{1}{\sqrt{\lambda_i}} A v_i \text{ where } i=1, \dots, d$$

$$\Rightarrow X u_i = A A^T \left(\frac{1}{\sqrt{\lambda_i}} A v_i \right) = \frac{1}{\sqrt{\lambda_i}} A (A^T A) v_i$$

$$= \frac{1}{\sqrt{\lambda_i}} A B v_i = \frac{1}{\sqrt{\lambda_i}} A \lambda_i v_i$$

$$= \sqrt{\lambda_i} A v_i = \lambda_i u_i$$

Thus u_i is indeed an eigenvector of X with eigenvalue λ_i .

So, the algorithm is as follows

Input:

$\Rightarrow$ Matrix $A \in \mathbb{R}^{m \times d}$ with $m \gg d$

$\Rightarrow$ Assume full rank & d distinct non-zero eigenvalues of $X = A A^T$

Output

$\Rightarrow$ Eigenvalues $\lambda_1, \dots, \lambda_d$

$\Rightarrow$ Eigenvectors $u_1, \dots, u_d \in \mathbb{R}^m$ of X

Steps:

$\Rightarrow$ Compute $B = A^T A \in \mathbb{R}^{d \times d}$

$\Rightarrow$ Compute eigendecomposition of B

$$B = V \Lambda V^T$$

$\Rightarrow$ Construct eigenvectors of $X = A A^T$
For each $i=1, \dots, d$:

$$\lambda_i = \Lambda_{i,i}$$

$v_i = \text{the } i^{\text{th}} \text{ column of } V$

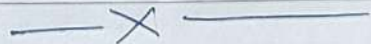
$$u_i = \frac{1}{\sqrt{\lambda_i}} A v_i \in \mathbb{R}^m$$

$\Rightarrow$ Return $\{\lambda_i, u_i\}$ for $i=1, \dots, d$.

This is highly efficient when $m \gg d$.



cello



2) a) True

Linear Regression is a supervised learning technique. It requires labeled data - that is, a set of input features X & a corresponding target variable y . The goal is to learn a function that maps inputs to outputs, minimizes prediction error.

2) b) True

PCA is an unsupervised learning technique. It does not use labeled data & is used to discover structure in the data. It helps in dimensionality reduction & visualizing high dimensional data.

2) c) False. The statement is reversed.

SVD is a matrix operation that factorizes any real matrix into three components $U \Sigma V^T$.

PCA is an operation on a dataset typically performed to extract principal directions of variance.

2) d) By using PCA as a preprocessing step.

To handle multicollinearity problem, apply PCA before regression to transform the original input features into a set of orthogonal (uncorrelated) components. Then we can use top- k features for dimensionality reduction if required.

